

Q65 - Approximation Vertex Cover Problem

Aim:

To compute a 2-approximation solution for the Vertex Cover problem.

Input:

Graph $G(V, E)$

Output:

Approximate Vertex Cover Set

Algorithm:

1. Select an uncovered edge (u, v) .
2. Add both u and v to the vertex cover set.
3. Remove all edges incident on u and v .
4. Repeat until no uncovered edges remain.

Approximation Logic:

The algorithm guarantees that the size of the generated cover is at most twice the size of the optimal solution.

Correctness:

Every selected edge is covered because both endpoints are included in the cover set.

Time Complexity:

$O(E)$

Approximation Ratio:

2-Approximation

Result:

An approximate vertex cover was successfully generated for the given graph.

Conclusion:

The approximation algorithm efficiently produces a near-optimal vertex cover for NP-Complete problems.

Output Screenshot:

(Insert terminal output screenshot here.)